

Re-write each expression into vertex form by completing the square.

1. $5x^2 - 20x + 6$

$$5(x^2 + 4x + 4) + 6 - 20$$

$$5(x - 2)^2 - 14$$

2. $3x^2 - 6x - 34$

$$3(x^2 - 2x + 1) - 34 - 3$$

$$3(x - 1)^2 - 37$$

3. $3y^2 - 6y - 3$

$$3(y^2 - 2y + 1) - 3 - 3$$

$$3(y - 1)^2 - 6$$

4. $2b^2 + 8b - 7$

$$2(b^2 + 4b + 4) - 7 - 8$$

$$2(b + 2)^2 - 15$$

5. $4p^2 + 16p - 15$

$$4(p^2 + 4p + 4) - 15 - 16$$

$$4(p + 2)^2 - 31$$

6. $9a^2 - 18a - 21$

$$9(a^2 - 2a + 1) - 21 - 9$$

$$9(a - 1)^2 - 30$$

7. $3r^2 - 12r + 7$
 $3(r^2 - 4r + 4) + 7 - 12$
 $3(r - 2)^2 - 5$

8. $3z^2 - 18z + 5$
 $3(z^2 - 6z + 9) + 5 - 27$
 $3(z - 3)^2 - 22$

9. $2t^2 - 12t - 1$
 $2(t^2 - 6t + 9) - 1 - 18$
 $2(t - 3)^2 - 19$

10. $3f^2 - 12f - 9$
 $3(f^2 - 4f + 4) - 9 - 12$
 $3(f - 2)^2 - 21$